
A Mathematical Illustration of Ambiguity and Vagueness in Textual Language

Benjamin Gardner*
School of Law
University of North Carolina
Chapel Hill, NC 27514
benjy@unc.edu

Abstract

This paper deals with failures of clarity in textual language when the text is syntactically and lexically correct: correct sentences that are unclear, usually by admitting multiple interpretations, rather than incorrect sentences. It identifies three such failures — implicitness, ambiguity, and vagueness — and shows that each corresponds to a precise mathematical object, drawing on Frege, Kant, and Wittgenstein to link text to formalism. The payoff is a claim about law: the interpretive problems at the base of legal practice are instances of a well-understood class of ill-posed inverse problems, and the discipline could describe — and partially engineer away — its own imprecision with tools it does not currently use.

1 Why is Clarity Important?

Language, broadly defined — statutes, computer code, large language model output — is a technology: a set of instructions. Clarity is the property that determines how precisely the instruction set is specified.

Consider the question "what is good legal writing?" As posed, it is not well defined. To sort legal texts into a "good" bin and a "not good" bin, one must first fix the ruler doing the sorting, and there are many candidate rulers: good if it supports my case, good if it is easy to understand, good if it earns an acceptable grade. The honest form of the judgment is "good (for my purposes)," and the parenthetical does real work. A text optimized for a professor's preferences and a text optimized for its author's enjoyment are different texts, and a text written for both will serve each purpose less well than a text written for either alone. The law student is therefore better served by "good legal writing *strategy*" — learn the grader's ruler and write to it — than by "good legal writing" simpliciter, and the same holds after graduation, when the ruler belongs to a judge.

These examples span human-to-human, human-to-machine, and machine-to-human instruction. The X-to-machine cases differ in one structural respect: the machine imposes a strict grammar and rejects violations. Upload an old Word document where a newer format is required and the system errors out; run Python containing non-code text and the interpreter throws. X-to-human language has no such gate. A human cannot ask "are you running version 1.4?" and thereby execute thousands of compatibility checks at once; the checks must be made piecemeal. "Are you a republican?" will somewhat clarify what is meant by "good policy."

The thesis of this paper is that the failures of clarity that slip through this missing gate — implicitness, ambiguity, and vagueness — are not mysteries. Each corresponds to a precise, well-understood

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

mathematical object, and a discipline built on text, as law is, could pose its interpretive problems with the same rigor.

2 Implicitness, Ambiguity, and Vagueness

Throughout, let D be a domain of objects and S a set of strings.

2.1 Implicitness

Definition 1 (Predicate). A predicate of arity n over D is a function $P : D^n \rightarrow \{0, 1\}$.²

Definition 2 (Saturation). An occurrence of an n -ary predicate is saturated when all n argument positions, and all contextual parameters on which its evaluation depends, are assigned values. Otherwise it is unsaturated.³

"Shorter" is binary: $\text{shorter}(\cdot, \cdot)$. The sentence "A is shorter" supplies one of its two arguments. In Frege's terminology the expression is unsaturated [3]: $\text{shorter}(A, \cdot)$ is a function awaiting an argument, not a proposition, and so it has no truth value. It is not false; it is incomplete.

A machine treats incompleteness as an error. Define `def shorter(a, b)` in Python and call `shorter(a)`, and the interpreter throws `TypeError: missing 1 required positional argument`. The human reader throws no error. She silently supplies the missing argument from context — the salient comparator ("shorter [than his brother, whom we were just discussing]"), a population default ("shorter [than the average man]"), or whatever is nearest to hand.

Definition 3 (Implicitness). A string s is implicit when it expresses an unsaturated predicate occurrence: s determines a term with free variables $x_1, \dots, x_k$, and only the pair (s, g) of the string with an assignment g of values to those variables is truth-evaluable.⁴

The elided material is not limited to overt grammatical slots. "Higher taxes are good" elides a beneficiary (*good for whom?*); "it is raining" elides a time and a place [6]; "A is tall" elides a comparison class. Nor is implicitness a defect. It is compression: "shorter than the average adult American man measured standing in the year of utterance" is exhausting, and elision works perfectly well so long as writer and reader share the same completion rule. The failures this paper cares about begin when they do not.

2.2 Ambiguity

Suppose now the string is fully saturated and is *still* unclear. Ambiguity is the case where one string corresponds to multiple complete propositions. The collision may be lexical ("he sat by the bank"), syntactic ("I saw the man with the telescope," which has two parse trees), or a matter of quantifier scope ("every student read a book": the same book, or one each?).

Definition 4 (Expression map). Let P be a set of truth-evaluable propositions. The expression map $\pi : P \rightarrow S$ sends a proposition to the string that expresses it. For $s \in S$, the fiber of s is the preimage $\pi^{-1}(s) = \{p \in P : \pi(p) = s\}$.⁵

Definition 5 (Ambiguity). A string s is ambiguous when $\pi^{-1}(s)$ contains propositions p_1, p_2 with different truth conditions. Equivalently: ambiguity is the failure of π to be injective at s , in a truth-relevant way.⁶

²Plainly: a predicate is a fill-in-the-blank sentence that becomes true or false once its blanks are filled with things. "What is shorter than what" has two blanks, so its arity is two.

³Plainly: saturated means every blank is filled. "A is shorter than B" is saturated; "A is shorter" still has an empty blank.

⁴Plainly: an implicit sentence leaves a blank for the reader to fill in from context — and expects her not to notice that she did.

⁵Plainly: π is the act of writing a thought down as words. The fiber of a sentence is the collection of every thought that would have come out as those exact words.

⁶Plainly: a sentence is ambiguous when two different thoughts — possibly one true and one false — would both have been written with those same words.

Figure 1 draws the definition. The string s_1 has a singleton fiber; the string s_2 is ambiguous, with three preimages, two true and one false. A reader handed s_2 knows only that the intended meaning lies somewhere in the dashed region.

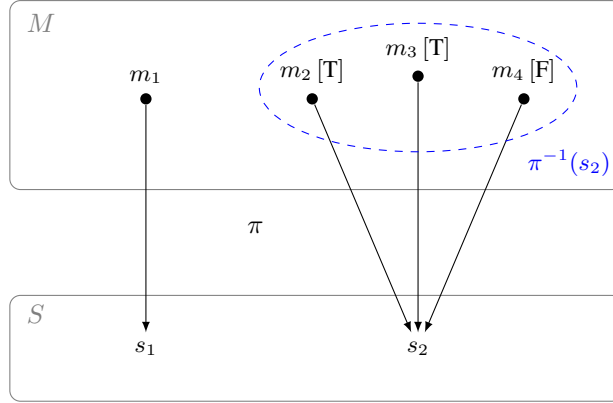


Figure 1: The expression map π from meanings M to strings S . The string s_1 is unambiguous: its fiber is a single proposition. The string s_2 is ambiguous: its fiber contains propositions of both truth values (Definition 5), and the string itself does not record which was meant.

Remark 1. *Implicitness and ambiguity are distinct pathologies. The implicit string is not yet a proposition — it is missing arguments: an under-application. The ambiguous string corresponds to too many propositions: a collision. Machine languages are engineered so that π , restricted to grammatical strings, is injective — a compiler guarantees a unique parse, and grammars that cannot guarantee one are rejected at design time. Natural language tolerates collisions because context usually, though not always, selects among them.*

2.3 Implicitness and Ambiguity Combined

The two pathologies compound. An elided slot with a single shared completion is harmless; an elided slot with multiple non-equivalent salient completions is an ambiguity smuggled in through an argument the reader did not notice she was supplying. "Higher taxes are good" is exactly this: the beneficiary slot is elided, and the candidate completions — for me, for the treasury, for long-run growth — yield different truth values.

The characteristic danger is *textual replication*. Because the completion lives in the context and not in the string, the string can be copied into a new context while the completion stays behind. The speaker meant "good (for me)"; the listener completes "good (for me)" with a different *me*; the sentence is repeated verbatim and its truth value silently flips in transit. Nothing in the text records that anything was lost, because the lost material was never in the text.

Communication succeeds when writer and reader select the same element of the fiber. The piecemeal checks of Section 1 are exactly this: "are you a republican?" acquires one coordinate of the speaker's context and thereby shrinks the set of completions the listener must consider.

2.4 Vagueness

There is a third failure, distinct from both. Suppose the string is saturated *and* has a unique reading: "A is tall for an American man." Every slot is filled and there is one parse — yet the sentence can lack a determinate truth value, because the extension of "tall" has no sharp boundary. If a man of 6'0" is tall, and a man one millimeter shorter than a tall man is tall, then by induction a man of 4'0" is tall. This is the sorites, and its lesson is that "tall" does not induce a partition of men into two bins.

Two formalizations are available, and they disagree about where the indeterminacy lives.

Definition 6 (Graded membership). *A fuzzy predicate over D replaces the characteristic function $1_P : D \rightarrow \{0, 1\}$ with a membership function $\mu_P : D \rightarrow [0, 1]$ [12]. Truth itself becomes graded.⁷*

⁷Plainly: instead of "tall" being yes-or-no, every height gets a score between 0 (clearly not tall) and 1 (clearly tall).

Definition 7 (Precisification and supertruth). A precisification of a predicate P is a classical sharpening $\hat{P} : D \rightarrow \{0, 1\}$ consistent with P 's clear cases (every admissible cutoff for "tall" between, say, 5'10" and 6'2"). A sentence is supertrue if true under all admissible precisifications, superfalse if false under all, and indeterminate otherwise [2]. Truth stays bivalent; the indeterminacy is relocated into language's failure to select one precisification.⁸

Remark 2. The two accounts make different commitments. Fuzzy semantics abandons bivalence and puts the blur in the truth value: "borderline-tall A is tall" is true to degree 0.6. Supervaluationism keeps every precisified world classical and puts the blur in the map from word to precisification: the sentence has no degree, only a status (supertrue, superfalse, indeterminate), and classical tautologies like " A is tall or A is not tall" remain supertrue even when neither disjunct is. Legal practice, without saying so, uses both: graded standards of proof ("preponderance," "clear and convincing") behave like thresholds on a $[0, 1]$ scale, while void-for-vagueness analysis asks the supervaluationist question of whether the admissible precisifications of a statute diverge on the defendant's conduct.

Figure 2 shows the two accounts side by side. Panel (a) is one curve with graded values; panel (b) is a family of classical step functions, and the shaded region — heights on which the admissible cutoffs disagree — is exactly the zone of indeterminacy.

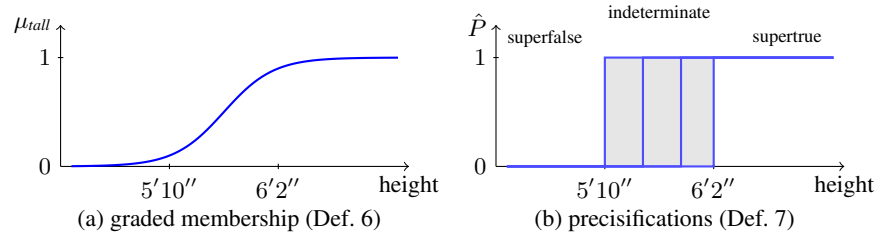


Figure 2: Two formalizations of "tall." (a) Fuzzy semantics: one membership function, graded truth. (b) Supervaluationism: a family of classical step functions, one per admissible cutoff; a sentence is supertrue where all agree on truth, superfalse where all agree on falsity, and indeterminate on the shaded band where they disagree.

Remark 3. On either account the contrast with ambiguity is the same. An ambiguous string points to a small discrete set of crisply different meanings; a vague string points to a single meaning surrounded by a continuum of near-identical precisifications. "Bank" has two meanings; "tall" has one blurry one. Vagueness also resists the repair that fixes implicitness: no finite list of necessary and sufficient conditions rescues "game," and Wittgenstein's family resemblances [11] suggest that for many predicates no such list exists to be elided in the first place.

3 Comparison: Context-Free and Environment-Indexed Relations

"Surgeons are doctors" (analytic) versus "surgeons are adults" (synthetic). In the analytic case only word meaning matters; in the synthetic case the world also matters. One can imagine a surgeon who is a child, but not a surgeon who is not a doctor.⁹ Synthetic-analytic confusion occurs when a speaker forgets the implicit "for me" in a value judgment such as "higher taxes are good (for me)": the context is absorbed into the statement, which can then be textually replicated in contexts where it is false.

Suppose person A is five feet tall and person B is six feet tall. The relation " A is shorter than B " is fixed by a characteristic of A (height), a characteristic of B (height), and the word's own meaning linking the two. A , in a vacuum, cannot be "shorter" — shorter than what? — and A cannot be shorter than an object C that lacks the relevant characteristic, e.g., an idea. Implicit in certain words, then, are other objects and specific characteristics of those objects; those objects and characteristics are synthetic to the word.

⁸Plainly: a precisification is one way of drawing a sharp line — say, "tall" means six feet and up. A sentence is supertrue if it stays true no matter where, within reason, the line is drawn.

⁹The analytic-synthetic distinction is itself contested [9]; the argument here needs only the weaker point that some truths of a sentence travel with its words and some do not.

Now define "better prospect" as "higher likelihood of greater future performance." Then A is better than B as a jockeying prospect and B is better than A as a basketball prospect. This is odd: how can A be simultaneously better and worse than B ? The earlier pathologies are absent — the word has valid objects and the objects have valid characteristics — yet " A is better than B as a prospect" appears simultaneously true and false.

One could repair the sentence by loading the context into the predicate as a further characteristic: "better basketball prospect." But characteristics chain without bound — "better college basketball prospect," "better SEC point-guard prospect in the 2027 draft class" — and any characteristic can itself be qualified. The cleaner move holds the objects and their intrinsic characteristics fixed and indexes the *relation*.

Definition 8 (Environment-indexed relation). *Let E be a set of environments. An environment-indexed relation on D is a family $\{R_e\}_{e \in E}$ of relations $R_e \subseteq D \times D$. A string that names the family without specifying e is unsaturated in e (Definition 2).¹⁰*

Proposition 1. *If $\{R_e\}$ is environment-indexed and there exist $e_1, e_2 \in E$ with $R_{e_1}(A, B)$ true and $R_{e_2}(A, B)$ false, then the bare string " $A R B$ " is truth-indeterminate: its fiber contains propositions of both truth values.*

Proof. The saturated propositions $R_{e_1}(A, B)$ and $R_{e_2}(A, B)$ both express as the bare string, so both lie in its fiber, and by hypothesis they differ in truth value. $\square$

There is no single relation *better*; there is the family $\{better_e\}$, and the bare string fails to specify the member. "Shorter," by contrast, is context-free: it is the constant family, the same relation in every environment.

Figures 3 and 4 make the contrast pictorial. In Figure 3, the "shorter" arrow is fixed entirely by the heights written at the nodes; no environment appears because none is needed. In Figure 4, the two panels share identical nodes, identical heights, and an identical "shorter" arrow, yet the "better" arrow points in opposite directions, because each panel (each grid color) is a different environment. The string " A is better than B " is consistent with both panels; the string " A is shorter than B " is consistent with only one direction, ever.

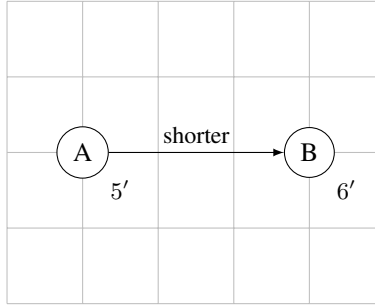


Figure 3: A context-free relation. The direction of "shorter" is determined by the characteristics of A and B alone; the same arrow appears in every environment.

4 Text as Projection

The pieces assemble into one geometric picture.

Definition 9 (Complete meaning). *A complete meaning consists of a core content c , an assignment g saturating its argument slots and parameters (§2.1), a parse t (§2.2), an environment e (§3), and a precisification $\hat{P}$ of each vague predicate (§2.4), collected as a tuple $m = (c, g, t, e, \hat{P})$.¹¹ Let M denote the space of complete meanings, and extend the expression map of Definition 4 to $\pi : M \rightarrow S$.*

¹⁰Plainly: some comparisons secretly depend on the setting. "Better" at basketball and "better" at horse racing are different comparisons wearing the same word.

¹¹Plainly: a complete meaning is the whole thought — the words, plus every blank filled, the setting named, and every line drawn.

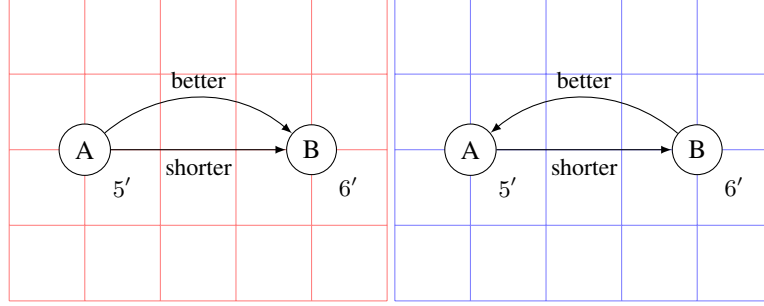


Figure 4: An environment-indexed relation. Left: the basketball environment. Right: the jockeying environment. Nodes, heights, and the "shorter" arrow are identical; only the environment (grid color) differs, and "better" reverses. The bare string " A is better than B " is true in one panel and false in the other (Proposition 1).

Writing is π , and π forgets coordinates: the assignment, the parse selection, the environment, and the precisification are components of the meaning that do not survive expression as text. Nothing in the string records what was forgotten — or that anything was.

Reading is the inverse problem: given the string, recover the meaning.

Proposition 2 (Ill-posedness of reading). *Reading is ill-posed in the sense of Hadamard [4]: a solution exists (the writer had a meaning) but is not unique whenever s is implicit (Definition 3), ambiguous (Definition 5), leaves the index of an environment-indexed relation unspecified (Proposition 1), or contains a vague predicate with multiple admissible precisifications (Definition 7). In each case $|\pi^{-1}(s)| > 1$.¹²*

The three failure modes are three shapes the fiber can take. Under benign implicitness, the fiber is large but writer and reader share a selection rule, so the reader lands on the intended element without noticing the fiber existed. Under ambiguity, the fiber contains elements with different truth values and no shared selection rule. Under vagueness, even the selected element is not a point but a band of precisifications.

Figure 5 makes the loss visible. Two distinct solids — a paraboloid and a cylinder — cast the identical shadow on the plane. An observer who receives only the shadow cannot distinguish them: the shadow is the text, the solids are the meanings, and the vertical dimension is the context that projection destroys.

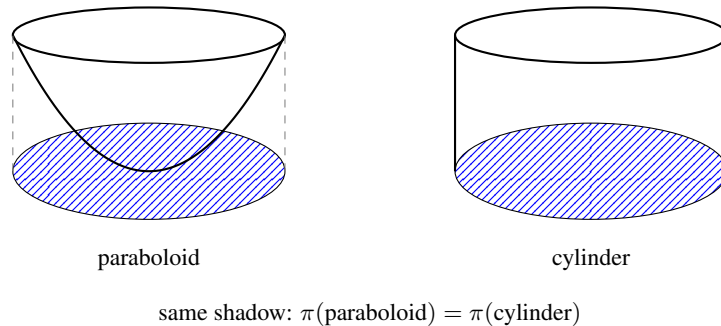


Figure 5: Distinct solids, one shadow. The paraboloid and the cylinder are different solids, yet each projects to the identical hatched ellipse in the plane. A reader in possession only of the shadow (the text) cannot recover which solid (which complete meaning) cast it.

¹²Plainly: an ill-posed puzzle is one where the clues you are given fit more than one answer.

5 Law as an Unregularized Inverse Problem

5.1 Regularization in Current Practice

Ill-posed inverse problems are solved, in practice, by *regularization*: the solver imposes prior structure that selects one candidate from the many consistent with the data [10].¹³ Reading works the same way. The reader carries a prior over M — over what people in this conversation, this genre, this era are likely to have meant — and selects the element of the fiber the prior favors. Communication succeeds when writer and reader regularize with the same prior.

Figure 6 depicts the mechanism. The fiber above a statutory string contains several candidate meanings; a shared prior over that fiber — however acquired — concentrates mass on one of them, and interpretation returns the selected element. The inverse problem is not solved; it is regularized.

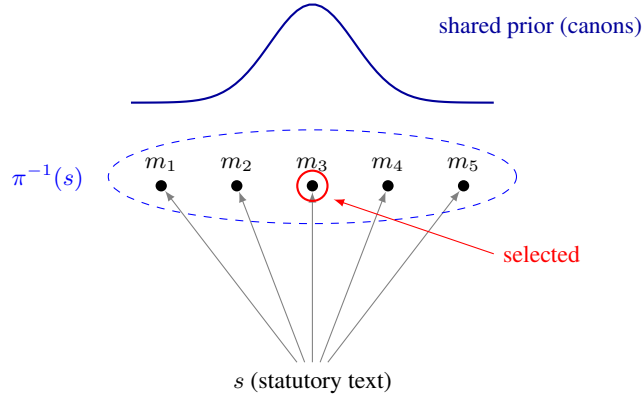


Figure 6: Interpretation as regularization. The statutory string s is consistent with the candidate meanings $m_1, \dots, m_5$ in its fiber. A shared prior — here supplied by the canons of construction — concentrates on m_3 , which interpretation returns. The prior does not make π injective; it makes the ill-posed inversion return a unique answer anyway.

Law has been regularizing all along without saying so. The canons of construction — *ejusdem generis*, *expressio unius*, the rule against surplusage — are published, shared selection rules over the fiber: given a statutory string consistent with many meanings, they tell every reader which coordinates to fill in and how. Statutory definitions sections are precisification-fixing devices (Definition 7): "for purposes of this chapter, 'employee' means ..." replaces a vague predicate with a stipulated sharp one over the statute's domain. Interpretive method disputes are, in this picture, disputes about the admissible coordinates of M : the textualist restricts the context a reader may consult to the enacted words and their contemporaneous public meaning, while the purposivist admits legislative history and statutory purpose as additional coordinates. Neither dissolves the ill-posedness (Proposition 2); they disagree about the space in which the problem is posed.

5.2 Sanity-Checking the Payoff

It is tempting to conclude that a sufficiently careful drafter could make π injective outright — saturate every slot, guarantee every parse, stipulate every boundary — and leave the interpreter a fiber of size one. Four limits scope that claim before it can be recommended.

Global saturation is combinatorially explosive. A statute is written once and applied to unboundedly many future situations. Enumerating every parameter of every predicate for every situation is the chaining problem of Section 3 in another guise: characteristics can be qualified without end, and the drafter cannot foresee the situations that will need them. This is Hart's open texture [5]: the rule "no vehicles in the park" was drafted against automobiles, and the skateboard, the ambulance, and the memorial tank arrive later. Some of the fiber's mass sits over worlds the drafter has not imagined, and no drafting effort spent today shrinks it.

¹³Plainly: when the clues fit several answers, a tiebreaker picks one. Regularization is the tiebreaker.

Some vagueness is deliberate. The literature on rules versus standards [7] makes the trade explicit: a precise rule is cheap to apply but must be specified before the facts are known, while a vague standard is costly to apply but lets the adjudicator use information available only after. "Reasonable care" is not a drafting failure; it is a delegation of precisification to the court, exercised at the moment when the facts are in. In the vocabulary of Section 3, the legislature transmits the family $\{R_e\}$ on purpose and assigns the choice of e to the judge.

Sharp cutoffs buy determinacy with arbitrariness and gaming. Precisify "tall" at 6'0" and the man at 5'11.9" has a grievance that the vague predicate never generated; precisify a tax threshold and the taxpayer restructures to land one dollar beneath it. The shaded band of Figure 2(b) does not vanish under stipulation; it compresses onto a single boundary where the stakes concentrate.

One limit is mathematical. For arbitrary context-free grammars, ambiguity is undecidable [1]: no algorithm can determine, for an arbitrary grammar, whether some string admits two parses. A regime that hopes to *test* for unique parses therefore cannot admit arbitrary legal English; it must draft within a grammar class that is unambiguous by construction, as programming languages do.

None of these limits defeats the thesis; they restate it. The achievable standard is not global injectivity but *chosen* imprecision: every string whose fiber exceeds one element should exceed it on purpose, with the excess accounted for.

5.3 Achieving the Three Standards Today

Each failure mode admits a concrete drafting discipline with existing precedent.

Saturation (against implicitness). The audit is mechanical: flag every relational or evaluative predicate in the bill and demand its slots — comparison class, beneficiary, environment, time — be filled or defaulted. Definitions sections already do half of this work; the other half is a checklist, and predicate–argument structure is exactly what a parser extracts, so the flagging can be automated.

Unique parse (against ambiguity). This standard is the most tractable, because syntactic ambiguity fails loudly and locally. It is also the most expensive to ignore: in *O'Connor v. Oakhurst Dairy*, a missing serial comma left a statutory list of exempt activities with two parses, the First Circuit read the fiber in the drivers' favor, and the ambiguity settled for roughly five million dollars.¹⁴ The remedy is a drafting grammar — restricted sentence forms, mandatory list punctuation — paired with an automatic parse check: if the parser returns two trees, the bill returns to the drafter. The rules-as-code movement has built the proof of concept: Catala [8] renders statutory logic as an executable, unique-parse artifact and has been validated against the French tax code.

Stipulation or delegation (against vagueness). For each load-bearing vague predicate, the drafter has two honest options. Stipulate the boundary where boundary arbitrariness is cheaper than case-by-case litigation — law already does this instinctively, replacing "reckless speed" with a posted limit and "intoxicated" with a numeric blood-alcohol threshold. Or delegate explicitly: name the precisifier (an agency, the court) and the admissible band within which the precisification must fall — the shaded region of Figure 2(b), written into the statute. The dishonest third option is the current default: silent vagueness, discovered in litigation and precisified retroactively at the litigants' expense.

5.4 A Statute for Statutes

Could a law require the three standards? Structurally, yes, and the model is the compiler. A drafting act would provide:

- (i) a published formal grammar for statutory text, drawn from a decidable, unambiguous-by-construction grammar class — the design move that evades the undecidability limit [1], exactly as programming-language designers evade it;
- (ii) a requirement that bills *compile*: an automated pre-enactment check that every sentence has exactly one parse under the grammar, and that every predicate's argument slots are saturated in place or defaulted by the definitions section;

¹⁴851 F.3d 69 (1st Cir. 2017). The disputed text exempted "packing for shipment or distribution": one parse exempts *distribution* as its own activity; the other exempts only *packing*, whether for shipment or for distribution. The drivers distributed but did not pack.

(iii) a *precisification schedule*: every predicate the check flags as vague either receives a stipulated boundary or an explicit delegation clause identifying the precisifier and the admissible band.

Compliance with all three is machine-testable before enactment, as a program is type-checked before it runs. What such a statute would guarantee is injectivity of π over the anticipated domain, plus a complete public accounting of the residue: every fiber of size greater than one is either eliminated or signed for. What it cannot guarantee is the unanticipated domain — open texture survives any grammar — and it cannot make delegation disappear, only visible.

That is the sane form of this paper's gripe. Law's imprecision is not eliminable, and some of it is load-bearing. But eliminable and deliberate imprecision are separable from accidental imprecision by tests that are a century old in mathematics and half a century old in compiler design, and law runs none of them. The foundational vocabulary lacks rigor not because rigor is unavailable, but because the discipline has not asked for it.

References

- [1] David G. Cantor. On the ambiguity problem of backus systems. *Journal of the ACM*, 9(4): 477–479, 1962.
- [2] Kit Fine. Vagueness, truth and logic. *Synthese*, 30:265–300, 1975.
- [3] Gottlob Frege. Über sinn und bedeutung. *Zeitschrift für Philosophie und philosophische Kritik*, 100:25–50, 1892.
- [4] Jacques Hadamard. Sur les problèmes aux dérivées partielles et leur signification physique. *Princeton University Bulletin*, 13:49–52, 1902.
- [5] H. L. A. Hart. *The Concept of Law*. Oxford University Press, 1961.
- [6] David Kaplan. Demonstratives. In Joseph Almog, John Perry, and Howard Wettstein, editors, *Themes from Kaplan*, pages 481–563. Oxford University Press, 1989.
- [7] Louis Kaplow. Rules versus standards: An economic analysis. *Duke Law Journal*, 42(3): 557–629, 1992.
- [8] Denis Mérigoux, Nicolas Chataing, and Jonathan Protzenko. Catala: A programming language for the law. *Proceedings of the ACM on Programming Languages*, 5(ICFP):1–29, 2021.
- [9] Willard Van Orman Quine. Two dogmas of empiricism. *The Philosophical Review*, 60(1): 20–43, 1951.
- [10] Andrey N. Tikhonov. Solution of incorrectly formulated problems and the regularization method. *Soviet Mathematics Doklady*, 4:1035–1038, 1963.
- [11] Ludwig Wittgenstein. *Philosophical Investigations*. Blackwell, 1953.
- [12] Lotfi A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.